

## Answers to exercise 02

### PART A

|    |   |    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|----|---|
| 1  | A | 2  | D | 3  | C | 4  | C | 5  | A |
| 6  | B | 7  | A | 8  | D | 9  | C | 10 | B |
| 11 | A | 12 | B | 13 | D | 14 | A | 15 | B |
| 16 | C | 17 | C | 18 | D | 19 | D | 20 | B |
| 21 | A | 22 | D | 23 | B | 24 | B | 25 | B |

### PART B

- (a) relation, (b) tuple, (c) attribute, (d) domain, (e) degree, (f) cardinality.
- (a) 24
- (a) distinct from / different from, (b) distinct, (c) the same, (d) distinct, (e) irrelevant, (f) irrelevant
- (a) uniquely identify, (b) super key, (c) candidate key, (d) primary key, (e) alternate keys, (f) foreign key (以上五小题答案与顺序无关).
- (a) candidate key, (b) super key, (c) candidate key, (d) candidate keys, (e) matches/maps/refers to candidate key
- (a) CID, (b) Student, (c) Class.
- (a) attribute, (b) primary key, (c) null
- (a) foreign key, (b) candidate key, (c) null, (d) appear/exist in
- (a) selection, (b) projection, (c) join, (d) division (本题答案与顺序无关).
- (a)  $R \cap S$ , (b)  $R \cup S$ , (c)  $R - S$
- (a) 2, (b) 1
- (a) Course, (b) Cno, (c) SC, (d) Sname='Zhangsan', (e) Student
- (a) Sno, (b) SC, (c) Student
- (a) Student, (b) Sno, grade, (c) SC, (d) Cname='database', (e) Course
- (a)  $(\prod_{A1, R, A2, A3} (\sigma_{R.A2=S.A2}(R \times S)))$

### PART C

- (a) Relation: a relation is uniquely named two-dimensional table with identically structured rows (technically tuples) and distinctly named columns (or attributes).
- (b) Attribute: attribute is a named column of a relation.
- (c) Domain: domain is a set of allowable values for attributes.
- (d) Tuple: tuple is a record/row of a relation.
- (e) Intension: the database schema is called the intension of the database.  
Extension: the database instance is called the extension of the database.
- (f) Degree: the number of attributes in a relation.  
Cardinality: the number of tuples in a relation.

### PART D

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Relations have six properties:

- (1) Homogeneity of the column/attribute
- (2) Different column can come from the same domain
- (3) The order in which columns appear in a relation is irrelevant
- (4) The order in which tuples appear in a relation is irrelevant
- (5) Each tuple/row should be different from others
- (6) Attribute is atomic, and could not be divided anymore